HK IMO Training 2023-2024
Problem Set 1
For 23 September 2023

Problem 1. There are n points on the plane which are not necessarily distinct. In each step, we can move two points to their midpoint. Our goal is to move all n points to the same point after a finite number of moves. Prove that we can meet our goal regardless of the initial positions of the n points if and only if n is a power of 2.

Problem 2. Does there exist a positive integer n such that

$$\pm 1^3 \pm 2^3 \pm 3^3 \pm \cdots \pm n^3 = 20162016$$

for some suitable choices of the signs?

Problem 3. Let S be a finite set of nonzero real numbers, and let $f : S \to S$ be a function. Suppose for every $x \in S$, we have $f(f(x)) = x + f(x)$ or $f(f(x)) = \dfrac{x + f(x)}{2}$. Prove that $f(x) = x$ for all $x \in S$.

Problem 4. In an acute $\triangle ABC$, let E and F be the feet of altitudes from B and C respectively, and let M be the midpoint of BC. The common external tangents to the circumcircles of $\triangle BEM$ and $\triangle CFM$ intersect at a point P. If P lies on the circumcircle of $\triangle ABC$, prove that $AP \perp BC$.